

Proper Title

Michał Wichrowski

Abstract

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Keywords: Keyords go here.

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1 Introduction

1.1 What *Has* Been Done (The Existing Literature)

If you dive into the literature on singularities in finite/hyperelasticity, you will mostly find two major phenomena:

- Crack Tips in Tension (Knowles and Sternberg, 1970s):** This is the most famous work on nonlinear singularities. They looked at a 360° corner (a crack) pulled in pure tension. They found that in a neo-Hookean material, the asymptotic displacement field behaves very differently from linear fracture mechanics. However, they only dealt with pure traction-free boundaries and pure tensile opening.
- Cavitation (John Ball, 1982):** Ball proved that under severe hydrostatic tension, hyperelastic equations admit singular weak solutions where a hole spontaneously opens in the middle of a perfectly solid domain (a point singularity where $\mathbf{F} \rightarrow \infty$).
- Surface Creasing (Biot 1963, Hong et al. 2009):** Biot found that if you compress a flat half-space of a neo-Hookean material, its surface becomes unstable at a critical stretch ($\lambda \approx 0.54$). Decades later, researchers realized this instability is a *crease* (a point singularity forming on a flat surface).

1.2 The Gap:

Almost all the existing nonlinear singularity literature avoids the exact setup you described: **a geometric corner (e.g., 90°) with mixed boundary conditions (Dirichlet meets Neumann) under compression.**

- Linear theory (Kondratiev) handles the mixed BC corner, but ignores the large-deformation kinematics.
- Nonlinear theory (Knowles/Sternberg) handles large deformations, but avoids the Dirichlet-Neumann condition and avoids compression.

⁰Interdisziplinäres Zentrum für Wissenschaftliches Rechnen (IWR), Ruprecht-Karls-Universität Heidelberg, Germany, mt.wichrowsk@uw.edu.pl

- Instability theory handles compression, but usually on flat domains or purely symmetric geometries, ignoring the existing geometric corner singularity.

The Gap: How does an already-existing stress concentration at a geometric Dirichlet-Neumann corner couple with a nonlinear structural instability (buckling/creasing)? How do we bound the operators there? That is fundamentally an open question.

2 Problem formulation

2.1 Finite-strain elastostatics as energy minimisation

We consider a hyperelastic body occupying a reference d -dimensional domain $\Omega \subset \mathbb{R}^d$ with boundary $\partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N}$, $\Gamma_D \cap \Gamma_N = \emptyset$. The deformation of the body is described by the displacement field $\mathbf{u} : \Omega \rightarrow \mathbb{R}^d$, the deformation map $\mathbf{y}(\mathbf{x}) = \mathbf{x} + \mathbf{u}(\mathbf{x})$, and the associated deformation gradient

$$\mathbf{F} = \nabla \mathbf{y} = \mathbf{I} + \nabla \mathbf{u}, \quad J = \det \mathbf{F} > 0, \quad (1)$$

where ∇ denotes the gradient with respect to the reference configuration. The mechanical response is encoded in a strain energy density $\Psi(\mathbf{F})$, and the equilibrium configuration is the minimiser of the total potential energy

$$\mathcal{E}(\mathbf{u}) = \int_{\Omega} \Psi(\mathbf{F}) dV - \int_{\Omega} \mathbf{f}_0 \cdot \mathbf{u} dV - \int_{\Gamma_N} \mathbf{t}_0 \cdot \mathbf{u} dS, \quad (2)$$

over the space of admissible displacements $U = \{\mathbf{u} \in H^1(\Omega; \mathbb{R}^2) : \mathbf{u} = \mathbf{g} \text{ on } \Gamma_D\}$, where $\mathbf{f}_0$ is a body force, $\mathbf{t}_0$ a prescribed traction on the Neumann boundary, and $\mathbf{g}$ the prescribed boundary displacement. We seek $\mathbf{u} \in U$ such that $\mathcal{E}(\mathbf{u}) \leq \mathcal{E}(\mathbf{v})$ for all admissible $\mathbf{v}$.

2.2 Stationarity, stress, and tangent modulus

A minimiser satisfies the stationarity condition $D_{\delta \mathbf{u}} \mathcal{E}(\mathbf{u}) = 0$ for all admissible variations $\delta \mathbf{u}$ (vanishing on Γ_D), which is the weak form of equilibrium

$$\int_{\Omega} \mathbf{P}(\mathbf{F}) : \nabla \delta \mathbf{u} dV = \int_{\Omega} \mathbf{f}_0 \cdot \delta \mathbf{u} dV + \int_{\Gamma_N} \mathbf{t}_0 \cdot \delta \mathbf{u} dS, \quad (3)$$

where the first Piola–Kirchhoff stress is the first derivative of the energy with respect to the deformation gradient,

$$\mathbf{P}(\mathbf{F}) = \frac{\partial \Psi}{\partial \mathbf{F}}. \quad (4)$$

The natural boundary condition associated with Γ_N is $\mathbf{P}\mathbf{N} = \mathbf{t}_0$, with $\mathbf{N}$ the reference outward unit normal. Because (3) is nonlinear in $\mathbf{u}$, it is solved iteratively by Newton’s method. Linearising the residual about a current iterate $\bar{\mathbf{u}}$ in a direction $\Delta \mathbf{u}$ introduces the fourth-order tangent (elasticity) modulus $\mathbb{L}$, with components L_{iJkL} ,

$$\mathbb{L} = \frac{\partial \mathbf{P}}{\partial \mathbf{F}} = \frac{\partial^2 \Psi}{\partial \mathbf{F} \otimes \partial \mathbf{F}}, \quad L_{iJkL} = \frac{\partial^2 \Psi}{\partial F_{iJ} \partial F_{kL}}, \quad (5)$$

which possesses the major symmetry $L_{iJkL} = L_{kLiJ}$, so that each Newton step requires solving the linear problem

$$\int_{\Omega} \nabla \delta \mathbf{u} : \mathbb{L}(\bar{\mathbf{F}}) : \nabla \Delta \mathbf{u} dV = -R(\bar{\mathbf{u}}; \delta \mathbf{u}), \quad (6)$$

with $R(\bar{\mathbf{u}}; \delta \mathbf{u})$ the residual of (3) evaluated at $\bar{\mathbf{u}}$. The well-posedness and conditioning of (6)—and hence the convergence of the whole nonlinear solver—are governed by the coercivity and boundedness of the bilinear form induced by $\mathbb{L}(\bar{\mathbf{F}})$. The central object of this work is therefore the behaviour of $\mathbb{L}$ as a function of the local deformation state.

2.3 The neo-Hookean material

To fix ideas and to exhibit explicit constants we use, as a running example, the compressible neo-Hookean strain energy density adopted in the cut-finite-element study [8] that motivates this analysis,

$$\Psi(\mathbf{F}) = \frac{\mu_e}{2} (\text{tr}(\mathbf{F}^T \mathbf{F}) - 2) + f(J), \quad (7)$$

where $\mu_e > 0$ is the shear modulus and $f(J)$ is a strictly convex volumetric penalty with $f(1) = 0$, $f'(1) = -\mu_e$, and $f(J) \rightarrow \infty$ as $J \rightarrow 0^+$. Two standard choices are

$$f_{\log}(J) = -\mu_e \log J + \lambda_e \log^2 J, \quad f_{\text{quad}}(J) = \frac{\kappa}{2} (J - 1)^2, \quad (8)$$

with λ_e the Lamé constant and κ a bulk modulus; (7) with $f_{\log}$ reproduces the model of [8]. The corresponding stress and tangent are

$$\mathbf{P}(\mathbf{F}) = \mu_e \mathbf{F} + f'(J) J \mathbf{F}^{-T}, \quad (9)$$

$$L_{iJkL} = \mu_e \delta_{ik} \delta_{JL} + (f''(J) J^2 + f'(J) J) F_{iL}^{-T} F_{kJ}^{-T} - f'(J) J F_{iJ}^{-T} F_{kL}^{-T}. \quad (10)$$

The first term $\mu_e \delta_{ik} \delta_{JL}$ is constant; the entire nonlinear part is controlled by J and the *inverse* deformation gradient $\mathbf{F}^{-1}$, yielding the structural bound

$$\|\mathbb{L}(\mathbf{F})\| \leq \mu_e + \Theta(J) \|\mathbf{F}^{-1}\|^2, \quad (11)$$

where $\|\mathbf{F}^{-1}\| = \lambda_{\min}^{-1}$ is set by the smallest principal stretch and Θ is a scalar growth function fixed by f (computed explicitly in §3.4). Consequently $\mathbb{L}$ can become unbounded through volumetric collapse, $\lambda_{\min} \rightarrow 0$ (equivalently $\mathbf{F}^{-1} \rightarrow \infty$); whether a purely tensile blow-up of $\mathbf{F}$ (with $\mathbf{F}^{-1}$ bounded) leaves $\mathbb{L}$ finite depends on the growth of $\Theta(J)$, which turns out to discriminate sharply between the two penalties in (8). This dichotomy is the analytical backbone of the present paper.

2.4 The mixed-corner singularity

The configuration of interest is a right-angle corner at the origin where the two boundary-condition types meet: a Dirichlet segment (Γ_D , prescribed displacement) abutting a traction-free Neumann segment (Γ_N). Such junctions arise, for example, at the lower corners of the loaded-pole benchmark of [8], where the numerically observed convergence rate is capped well below the optimal order. Classical Kondratiev theory for the linearised problem predicts a corner singularity

$$\mathbf{u}(r, \theta) \sim K r^\alpha \boldsymbol{\Phi}(\theta), \quad \alpha \in (0, 1), \quad (12)$$

in polar coordinates (r, θ) centred at the corner, with K a generalised stress-intensity factor and α the (geometry- and Poisson-ratio-dependent) singularity exponent. The deformation gradient then scales as $\mathbf{F} = \mathbf{I} + \mathcal{O}(K r^{\alpha-1})$, so $\|\mathbf{F}\| \rightarrow \infty$ as $r \rightarrow 0$.

The question this raises—and which the remainder of the paper addresses—is how this kinematic singularity propagates into the nonlinear tangent operator (5). In view of (11), the answer hinges on which principal stretch diverges: a tensile corner keeps $\mathbf{F}^{-1}$ bounded and (for a suitable

penalty) leaves $\mathbb{L}$ well behaved, whereas a compressive corner drives $\lambda_{\min} \rightarrow 0$ and blows up the *norm* of $\mathbb{L}$ (severe ill-conditioning). The tangent nonetheless remains strongly elliptic — the compressive instability is *structural* (a corner analogue of creasing), not a change of type of the PDE. The following sections analyse the two scenarios in turn (Section 3) and then show how a small imperfection regularises the compressive case through a structural bifurcation (Section 4).

3 Asymptotic behaviour of the tangent operator at a mixed corner

3.1 Geometry and linearised kinematics

Let the reference domain $\Omega \subset \mathbb{R}^2$ feature a right-angle corner at the origin $(0,0)$, described in polar coordinates (r, θ) by $r \in [0, R]$, $\theta \in [0, \pi/2]$, and write $B_\rho := \{\mathbf{x} \in \Omega : |\mathbf{x}| < \rho\}$ for the corner neighbourhood.

- **Dirichlet boundary** (Γ_D): $\theta = 0$, deformation prescribed, $\mathbf{y}(r, 0) = \mathbf{g}(r)$.
- **Neumann boundary** (Γ_N): $\theta = \pi/2$, traction-free, $\mathbf{P}\mathbf{N} = \mathbf{0}$.

Classical Kondratiev theory for the *linearised* problem gives the leading-order singular displacement

$$\mathbf{u}(r, \theta) \sim K r^\alpha \Phi(\theta), \quad (13)$$

with K the generalised stress-intensity factor (set by the far field), Φ a smooth angular profile, and $\alpha \in (0, 1)$ the geometry- and Poisson-dependent exponent. Consequently

$$\mathbf{F} = \mathbf{I} + \mathcal{O}(K r^{\alpha-1}), \quad \|\mathbf{F}\| \rightarrow \infty \text{ as } r \rightarrow 0. \quad (14)$$

Whether this kinematic blow-up contaminates the tangent operator $\mathbb{L} = \partial^2 \Psi / \partial \mathbf{F} \partial \mathbf{F}$ depends not on $\|\mathbf{F}\|$ but on which principal stretches diverge, which is the content of the rest of this section.

3.2 Standing assumptions on the stored energy

The mechanism we describe is not specific to the neo-Hookean model; that model only serves to fix ideas and to exhibit explicit constants. We therefore phrase the analysis for a general frame-indifferent energy and isolate the structural properties that the argument actually uses. Throughout, $\|\cdot\|$ denotes the operator (spectral) norm on tensors, $0 < \lambda_{\min}(\mathbf{F}) \leq \lambda_{\max}(\mathbf{F})$ are the singular values of $\mathbf{F}$ (the principal stretches), and we recall $\|\mathbf{F}^{-1}\| = \lambda_{\min}(\mathbf{F})^{-1}$.

Assumption 1 (Regularity and frame indifference). The stored energy $\Psi : \text{GL}^+(2) \rightarrow \mathbb{R}$ is objective, $\Psi \in C^2(\text{GL}^+(2))$, and $\Psi(\mathbf{F}) \rightarrow +\infty$ as $\det \mathbf{F} \rightarrow 0^+$.

Assumption 2 (Inverse-channel growth of the tangent). There exist a continuous, nondecreasing function $\Theta : (0, \infty) \rightarrow (0, \infty)$ and a constant $c_0 \geq 0$ such that the tangent modulus $\mathbb{L}(\mathbf{F}) = D^2 \Psi(\mathbf{F})$ satisfies, for all $\mathbf{F} \in \text{GL}^+(2)$,

$$\|\mathbb{L}(\mathbf{F})\| \leq c_0 + \Theta(\det \mathbf{F}) \|\mathbf{F}^{-1}\|^2. \quad (15)$$

Assumption 3 (Uniform strong ellipticity on bounded-distortion sets). For every $0 < c \leq 1$ and $M \geq 1$ there is $\gamma(c, M) > 0$ such that whenever $\lambda_{\min}(\mathbf{F}) \geq c$ and $\det \mathbf{F} \leq M$,

$$L_{iJkL}(\mathbf{F}) m_i N_J m_k N_L \geq \gamma(c, M) |\mathbf{m}|^2 |\mathbf{N}|^2 \quad \forall \mathbf{m}, \mathbf{N} \in \mathbb{R}^2. \quad (16)$$

The decisive feature of Assumption 2 is that the inverse-stretch factor $\|\mathbf{F}^{-1}\|^2 = \lambda_{\min}^{-2}$ is the *only* algebraic channel through which the Hessian can blow up; the dependence on $\|\mathbf{F}\|$ (equivalently on $\lambda_{\max}$) enters solely through the scalar $\Theta(\det \mathbf{F})$. Whether a tensile corner ($\lambda_{\max} \rightarrow \infty$, $\det \mathbf{F} \rightarrow \infty$, $\lambda_{\min}$ bounded below) is benign therefore reduces to the *growth rate* of Θ , which we quantify in §3.4. Materials with genuine tensile stiffening of the deviatoric part (Gent, Arruda–Boyce), whose tangent diverges through $\lambda_{\max}$ directly rather than through $\Theta(\det \mathbf{F})$, violate (15) and are excluded.

Lemma 1 (Assumptions for the neo-Hookean model). *Let Ψ be the neo-Hookean energy (7) with $\mu_e > 0$ and $f \in C^2(0, \infty)$. Then Assumption 2 holds with $c_0 = \mu_e$ and*

$$\Theta(J) = |f''(J)J^2 + f'(J)J| + |f'(J)J|, \quad (17)$$

and Assumption 3 holds (rank-one convexity away from $\det \mathbf{F} = 0$). Assumption 1 additionally requires the barrier property $f(J) \rightarrow \infty$ as $J \rightarrow 0^+$: this holds for $f_{\log}$ but fails for f_{quad} , since $f_{\text{quad}}(J) \rightarrow \frac{\kappa}{2} < \infty$ (see Remark 3 and Scenario B).

Proof. Objectivity and C^2 regularity on $\text{GL}^+(2)$ are immediate; the barrier hypothesis $f(J) \rightarrow \infty$ as $J \rightarrow 0^+$, when assumed, gives the blow-up required by Assumption 1. In (10) the first term is the scaled identity, of norm μ_e . Each of the remaining two terms is a rank-one, respectively transposition, contraction of $\mathbf{F}^{-T} \otimes \mathbf{F}^{-T}$; since $\|\mathbf{F}^{-T}\| = \|\mathbf{F}^{-1}\|$, each is bounded in operator norm by its scalar prefactor times $\|\mathbf{F}^{-1}\|^2$. The triangle inequality gives (15) with $c_0 = \mu_e$ and Θ as in (17). For Assumption 3, contracting (10) with $\mathbf{m} \otimes \mathbf{N}$ and writing $\xi := \mathbf{m} \cdot \mathbf{F}^{-T} \mathbf{N} = \mathbf{N} \cdot \mathbf{F}^{-1} \mathbf{m}$, the two volumetric terms combine $((f''J^2 + f'J)\xi^2 - f'J\xi^2)$ to the single clean form

$$L_{iJkL}m_iN_Jm_kN_L = \mu_e |\mathbf{m}|^2 |\mathbf{N}|^2 + f''(J) J^2 \xi^2. \quad (18)$$

Wherever $f''(J) \geq 0$ the second term is nonnegative and (16) holds with $\gamma = \mu_e$, uniformly in $\mathbf{F}$ (no distortion bound needed); this covers f_{quad} ($f'' = \kappa > 0$) everywhere and $f_{\log}$ on $\{f''_{\log} \geq 0\}$. $\square$ $\square$

Remark 1 (Strength and a tension caveat). Identity (18) is sharper than (16): the neo-Hookean tangent is strongly elliptic with constant *exactly* μ_e wherever $f''(J) \geq 0$, and the volumetric term only adds to it. The Hessian of the quadratic (Hookean) part is constant, which is why the *deviatoric* response never blows up; only the *volumetric* response, through Θ , can. One caveat for $f_{\log}$: since $f''_{\log}(J) = (\mu_e + 2\lambda_e - 2\lambda_e \log J)/J^2$, convexity is lost once $J > J_* := \exp(1 + \mu_e/(2\lambda_e))$, and then (18) can turn negative for ξ near its maximal value $|\mathbf{m}||\mathbf{N}|/\lambda_{\min}$. Hence Assumption 3 holds throughout *compression* ($J \leq 1$) and moderate tension, but $f_{\log}$ can lose strong ellipticity under *extreme dilation* ($J > J_*$)—a tension-side phenomenon distinct from the compressive analysis, and one more reason the bounded- J qualifier matters in §3.5. The globally convex f_{quad} is free of this.

3.3 The non-compressive hypothesis

The qualitative split between the two scenarios is captured by a single quantity: the smallest principal stretch near the corner. We call the corner singularity *non-compressive* when $\lambda_{\min}$ stays bounded away from zero, even though $\lambda_{\max}$ (hence $\|\mathbf{F}\|$ and $\det \mathbf{F}$) may diverge.

Assumption 4 (Non-compressive singularity, (NC)). There exist $\rho \in (0, R]$ and $c \in (0, 1]$ such that the equilibrium deformation satisfies

$$\lambda_{\min}(\mathbf{F}(\mathbf{x})) \geq c, \quad \text{equivalently} \quad \|\mathbf{F}^{-1}(\mathbf{x})\| \leq c^{-1}, \quad \text{for a.e. } \mathbf{x} \in B_\rho. \quad (19)$$

We stress what (19) does *not* say. It does not require incompressibility, and crucially it does *not* bound $\det \mathbf{F}$: in a tensile corner $\lambda_{\max} \sim r^{\alpha-1} \rightarrow \infty$, so $J = \lambda_{\min} \lambda_{\max} \sim r^{\alpha-1} \rightarrow \infty$ as well. The only quantity (NC) controls is $\|\mathbf{F}^{-1}\| \leq c^{-1}$. Physically, (NC) states that the material is pulled around the corner (tension/shear) rather than crushed into it. Because $\det \mathbf{F}$ is *not* bounded under (NC), the factor $\Theta(\det \mathbf{F})$ in (15) need not stay bounded; the next subsection makes its growth explicit, and this is precisely where the two volumetric penalties (8) part ways. Scenario B (§3.6) treats the complementary, compressive case in which (19) fails.

3.4 Explicit volumetric growth functions

The bound (15) exposes *two independent channels* through which the tangent can diverge, and they are kinematically decoupled:

$$\|\mathbb{L}(\mathbf{F})\| \leq \mu_e + \underbrace{\Theta(\det \mathbf{F})}_{\substack{\text{dilation channel} \\ J \rightarrow \infty}} \cdot \underbrace{\|\mathbf{F}^{-1}\|^2}_{\substack{\text{inverse channel} \\ \lambda_{\min} \rightarrow 0}}. \quad (20)$$

The *inverse channel* is the one usually blamed for trouble: it requires $\lambda_{\min} \rightarrow 0$, i.e. compression to zero volume, and is the subject of Scenario B. The *dilation channel*, $\Theta(\det \mathbf{F}) \rightarrow \infty$ as $J \rightarrow \infty$, is activated by *expansion* and is precisely what a tensile corner switches on, since $\lambda_{\max} \sim r^{\alpha-1} \rightarrow \infty$ forces $J \rightarrow \infty$ even with $\lambda_{\min}$ bounded below by (NC). Whether this matters is therefore not a property of the corner or of the sign of the loading, but of the *growth rate of Θ* —a property of the volumetric penalty f alone. We now evaluate it for the two penalties (8); the contrast shows that “tension is benign” is a model-dependent statement.

Logarithmic penalty. For $f_{\log}(J) = -\mu_e \log J + \lambda_e \log^2 J$,

$$f'(J)J = -\mu_e + 2\lambda_e \log J, \quad f''(J)J^2 = \mu_e + 2\lambda_e - 2\lambda_e \log J, \quad (21)$$

so that the combination collapses to a constant,

$$f''(J)J^2 + f'(J)J = 2\lambda_e, \quad (22)$$

and therefore

$$\Theta_{\log}(J) = 2\lambda_e + |2\lambda_e \log J - \mu_e| = \mathcal{O}(\log J) \quad \text{as } J \rightarrow \infty. \quad (23)$$

The volumetric stiffening grows only *logarithmically* in J .

Quadratic penalty. For $f_{\text{quad}}(J) = \frac{\kappa}{2}(J-1)^2$,

$$f'(J)J = \kappa J(J-1), \quad f''(J)J^2 = \kappa J^2, \quad f''(J)J^2 + f'(J)J = \kappa J(2J-1), \quad (24)$$

so

$$\Theta_{\text{quad}}(J) = \kappa J |2J-1| + \kappa J |J-1| = \mathcal{O}(\kappa J^2) \quad \text{as } J \rightarrow \infty. \quad (25)$$

The volumetric stiffening grows *quadratically* in J .

Consequence at the corner. We combine (15) with (NC) ($\|\mathbf{F}^{-1}\|^2 \leq c^{-2}$) and the tensile kinematics. In the non-compressive regime exactly one stretch diverges: $\lambda_{\max} \sim |K| r^{\alpha-1}$ while $\lambda_{\min}$ stays bounded below, so $J = \lambda_{\min} \lambda_{\max} \sim r^{\alpha-1}$. (If instead $\nabla \mathbf{u}$ had two diverging eigenvalues, the subdeterminant would give $J \sim r^{2(\alpha-1)}$; we work in the one-large-stretch regime characteristic of a Dirichlet–Neumann opening, where $J \sim r^{\alpha-1}$.) The tangent norm near the corner then obeys

$$\|\mathbb{L}(\mathbf{x})\| \leq \mu_e + c^{-2} \Theta(\det \mathbf{F}(\mathbf{x})) \sim \begin{cases} \mu_e + \mathcal{O}(\log \frac{1}{r}), & f = f_{\log}, \\ \mathcal{O}(r^{-2(1-\alpha)}), & f = f_{\text{quad}}. \end{cases} \quad (26)$$

Thus the logarithmic penalty keeps the tangent bounded up to a logarithm, whereas the quadratic penalty produces a genuine *algebraic* blow-up of the tangent *even in pure tension*. This is the dilation channel of (20) acting alone, with the inverse channel inert ($\|\mathbf{F}^{-1}\|$ bounded); it is algebraically singular of a severity comparable to—though mechanistically distinct from—the compressive blow-up of Scenario B, which instead runs through $\|\mathbf{F}^{-1}\|^2$. The often-repeated statement that “tension is harmless for neo-Hookean” is therefore an artefact of the logarithmic volumetric model and is false for the quadratic one.

Physical reading: the large-dilation response. The dichotomy is, at bottom, a statement about how the bulk energy behaves under unbounded *expansion*, the loading a tensile corner imposes:

volumetric model	energy as $J \rightarrow \infty$	tangent $\Theta(J)$
$\lambda_e \log^2 J$	$\sim \log^2 J$ (soft)	$\mathcal{O}(\log J)$
$\frac{\kappa}{2}(J-1)^2$	$\sim J^2$ (stiff)	$\mathcal{O}(J^2)$

A penalty that is polynomial in J makes the material resist further dilation ever more stiffly, and the linearised modulus faithfully reports that stiffness as a blow-up; the logarithmic penalty is gentle about expansion and does not. Both penalties are designed to diverge correctly as $J \rightarrow 0$ (resisting collapse, the inverse channel); their large- J behaviour, which the dilation channel probes, was never physically constrained, and that latitude is exactly where they part. The exact cancellation (22) is the analytic signature of this softness. We make the benign case precise in Theorem 1 and flag the algebraic case in Remark 3.

3.5 Scenario A: the tensile (non-compressive) singularity

We work under the mild-growth condition isolated above.

Assumption 5 (Mild volumetric growth). The growth function satisfies $\Theta(J) = \mathcal{O}((\log J)^p)$ as $J \rightarrow \infty$ for some $p \geq 0$.

By (23), $f_{\log}$ satisfies Assumption 5 with $p = 1$; by (25), f_{quad} does not.

Lemma 2 (Subordinate tangent). *Under Assumptions 1, 2, 5 and (NC), for every $\varepsilon > 0$ there is C_ε with*

$$\|\mathbb{L}(\mathbf{x})\| \leq C_\varepsilon r^{-\varepsilon} \quad \text{for a.e. } \mathbf{x} \in B_\rho. \quad (27)$$

In particular $\mathbb{L} \in L^q(B_\rho)$ for every $q < \infty$, and $\|\mathbb{L}\|$ is dominated by any negative power of r .

Proof. By (15) and (NC), $\|\mathbb{L}\| \leq \mu_e + c^{-2} \Theta(\det \mathbf{F})$. Assumption 5 and $\det \mathbf{F} = \mathcal{O}(r^{\alpha-1})$ give $\Theta(\det \mathbf{F}) = \mathcal{O}((\log \frac{1}{r})^p)$, and $(\log \frac{1}{r})^p \leq C_\varepsilon r^{-\varepsilon}$ on $(0, \rho]$ for every $\varepsilon > 0$. $\square$ $\square$

Lemma 3 (Uniform ellipticity). *Under Assumptions 1, 3 and (NC), and assuming the volumetric prefactors in (18) remain controlled (automatic for $f_{\log}$, where $f''J^2 + f'J = 2\lambda_e$ is constant and $|f'J| = \mathcal{O}(\log J)$ is subordinate), the tangent satisfies the Legendre–Hadamard condition on B_ρ with a slowly varying constant: there are $\gamma_0 > 0$ and $\varepsilon > 0$ arbitrarily small with*

$$L_{iJkL}(\mathbf{x}) m_i N_J m_k N_L \geq (\gamma_0 - C_\varepsilon r^\varepsilon) |\mathbf{m}|^2 |\mathbf{N}|^2 \geq \frac{1}{2} \gamma_0 |\mathbf{m}|^2 |\mathbf{N}|^2 \quad (28)$$

for a.e. $\mathbf{x} \in B_\rho$ after shrinking ρ , for all $\mathbf{m}, \mathbf{N} \in \mathbb{R}^2$.

Proof. From (18), $L_{iJkL} m_i N_J m_k N_L \geq \mu_e |\mathbf{m}|^2 |\mathbf{N}|^2 - |f''J^2 + f'J| |\mathbf{a}|^2 |\mathbf{m}|^2 - |f'J| |\mathbf{a}| |\mathbf{b}| |\mathbf{m}| |\mathbf{N}|$, where the volumetric correction is bounded using $|\mathbf{a}| \leq c^{-1} |\mathbf{N}|$, $|\mathbf{b}| \leq c^{-1} |\mathbf{m}|$ by $c^{-2} (|f''J^2 + f'J| + |f'J|) |\mathbf{m}|^2 |\mathbf{N}|^2 = c^{-2} \Theta(J) |\mathbf{m}|^2 |\mathbf{N}|^2$. For $f_{\log}$, $\Theta(J) = \mathcal{O}(\log \frac{1}{r}) \leq C_\varepsilon r^\varepsilon$ on $(0, \rho]$; the volumetric correction is thus $o(1)$ as $r \rightarrow 0$, and after shrinking ρ it is below $\frac{1}{2} \mu_e$, giving (28) with $\gamma_0 = \mu_e$. $\square$ $\square$

For the weighted well-posedness we recall two standard inequalities in the Kondratiev space $V_\beta^{1,2}(B_\rho)$ —the closure of smooth functions vanishing on $\Gamma_D \cap \partial B_\rho$ under $\|\mathbf{v}\|_{V_\beta^{1,2}}^2 = \int_{B_\rho} (r^{2\beta} |\nabla \mathbf{v}|^2 + r^{2\beta-2} |\mathbf{v}|^2) dV$.

Lemma 4 (Weighted Hardy and Korn inequalities [6, 7]). *Let $\beta \in \mathbb{R}$ avoid the (discrete) set of exceptional exponents of the corner. There are constants C_H, C_K depending only on β and the opening angle such that, for all $\mathbf{v} \in V_\beta^{1,2}(B_\rho)$ vanishing on $\Gamma_D \cap \partial B_\rho$,*

$$\int_{B_\rho} r^{2\beta-2} |\mathbf{v}|^2 dV \leq C_H \int_{B_\rho} r^{2\beta} |\nabla \mathbf{v}|^2 dV, \quad (29)$$

$$\int_{B_\rho} r^{2\beta} |\nabla \mathbf{v}|^2 dV \leq C_K \int_{B_\rho} r^{2\beta} |\boldsymbol{\varepsilon}(\mathbf{v})|^2 dV, \quad \boldsymbol{\varepsilon}(\mathbf{v}) = \frac{1}{2} (\nabla \mathbf{v} + \nabla \mathbf{v}^T). \quad (30)$$

Let $\beta_\star$ be the weight exponent for which the linear elasticity corner problem at this Dirichlet–Neumann junction is Fredholm (equivalently $\beta_\star$ avoids the singular exponents α). We may write the Newton-step bilinear form as

$$a(\Delta \mathbf{u}, \delta \mathbf{u}) := \int_{B_\rho} \nabla \delta \mathbf{u} : \mathbb{L}(\bar{\mathbf{F}}) : \nabla \Delta \mathbf{u} dV. \quad (31)$$

Theorem 1 (Well-posedness of the linearised step, tensile corner). *Under Assumptions 1–5 and the non-compressive hypothesis (NC), the form (31) is bounded and coercive on $V_{\beta_\star}^{1,2}(B_\rho)$ modulo a compact perturbation: there exist constants $\Lambda < \infty$, $\gamma_1 > 0$ and a compact form $k(\cdot, \cdot)$ such that, for all $\mathbf{u}, \mathbf{v} \in V_{\beta_\star}^{1,2}(B_\rho)$,*

$$|a(\mathbf{u}, \mathbf{v})| \leq \Lambda \|\mathbf{u}\|_{V_{\beta_\star}^{1,2}} \|\mathbf{v}\|_{V_{\beta_\star}^{1,2}}, \quad (32)$$

$$a(\mathbf{v}, \mathbf{v}) \geq \gamma_1 \|\mathbf{v}\|_{V_{\beta_\star}^{1,2}}^2 - k(\mathbf{v}, \mathbf{v}). \quad (33)$$

Consequently the linearised corner operator is Fredholm of index zero in $V_{\beta_\star}^{1,2}$; for $\beta_\star$ avoiding the corner spectrum it is an isomorphism, its solutions admit the asymptotic expansion (13), and the governing singular exponents coincide with those of the linearised elasticity problem.

Proof. Boundedness. By Cauchy–Schwarz and Lemma 2, for any $\varepsilon \in (0, 1)$,

$$\begin{aligned} |a(\mathbf{u}, \mathbf{v})| &\leq \int_{B_\rho} \|\mathbb{L}\| |\nabla \mathbf{u}| |\nabla \mathbf{v}| dV \leq C_\varepsilon \int_{B_\rho} r^{-\varepsilon} |\nabla \mathbf{u}| |\nabla \mathbf{v}| dV \\ &\leq C_\varepsilon \left(\int_{B_\rho} r^{2\beta_\star} |\nabla \mathbf{u}|^2 \right)^{1/2} \left(\int_{B_\rho} r^{-2\varepsilon-2\beta_\star} |\nabla \mathbf{v}|^2 \right)^{1/2}. \end{aligned}$$

Since ε is arbitrary and $\beta_\star$ is interior to an interval of admissible weights, the loss $r^{-2\varepsilon}$ can be absorbed by a weight chosen within that interval; this gives (32) with $\Lambda = \Lambda(\varepsilon)$. The key point is that the mild (logarithmic) growth of $\|\mathbb{L}\|$ is subordinate to every algebraic weight, so it does not change the admissible weight interval.

Gårding inequality. By Lemma 3 the integrand symbol $\mathbb{L}(\mathbf{x})$ is uniformly strongly elliptic on B_ρ with constant $\geq \frac{1}{2}\gamma_0$. Freezing the (measurable, uniformly elliptic) coefficients on a dyadic partition $\{r \sim 2^{-j}\}$ and applying the constant-coefficient Gårding inequality on each annulus—valid because the principal symbol is positive definite on the Legendre–Hadamard cone—yields, after summation against the weight $r^{2\beta_\star}$,

$$a(\mathbf{v}, \mathbf{v}) \geq \frac{1}{2}\gamma_0 c_K^{-1} \int_{B_\rho} r^{2\beta_\star} |\nabla \mathbf{v}|^2 dV - C \int_{B_\rho} r^{2\beta_\star} |\mathbf{v}|^2 dV,$$

where the Korn inequality (30) converts ellipticity of the symmetric part into control of the full gradient. The remainder term has a strictly higher power of r than the Hardy weight $r^{2\beta_\star-2}$, hence by (29) it is relatively compact with respect to $\|\cdot\|_{V_{\beta_\star}^{1,2}}$; this is the compact form k , and we obtain (33) with $\gamma_1 = \frac{1}{2}\gamma_0 c_K^{-1}$.

Fredholm property and exponents. Estimates (32)–(33) place the operator in the Kondratiev–Maz’ya framework [6]: it is Fredholm of index zero in $V_{\beta_\star}^{1,2}$ whenever $\beta_\star$ avoids the spectrum of the corner operator pencil. Because the nonlinear correction to the principal symbol is subordinate (it is $o(r^{-\varepsilon})$ for every ε , by Lemma 2), the pencil coincides with that of linearised elasticity; its eigenvalues are exactly the exponents α of (13), which are therefore unchanged by the nonlinearity. $\square$ $\square$

Remark 2 (Stress is singular, operator is not). The first Piola–Kirchhoff stress still blows up, $\mathbf{P} \sim \mu_e \mathbf{F} \rightarrow \infty$ as $r \rightarrow 0$, so the tensile corner is a genuine *stress* singularity. Theorem 1 asserts only that this singularity lives entirely in the *right-hand side* (the residual) of the Newton step, not in its operator: under a mild volumetric penalty the nonlinear problem inherits the regularity theory of the linear one, standard graded meshes restore optimal convergence, and no structural instability occurs. This is the precise sense in which a non-compressive corner “passes like linear elasticity.”

Remark 3 (The quadratic penalty does *not* pass like linear elasticity). For f_{quad} , Assumption 5 fails: by (25) and (26) the tangent grows algebraically, $\|\mathbb{L}\| = \mathcal{O}(r^{-2(1-\alpha)})$, *even in pure tension*. The coefficients of (31) are then genuinely singular, of severity comparable to the compressive Scenario B, the boundedness step above fails for $\beta_\star$, and the admissible weight interval—hence the effective corner exponents seen by the nonlinear operator—shifts relative to linear elasticity. A sharp analysis of this case (the correct weight, and whether coercivity survives) is governed by the volumetric blow-up rather than by the deviatoric singularity and is left open here.

We emphasise that this serves as a *cautionary comparator* rather than a model we advocate: as written, $f'_{\text{quad}}(1) = 0 \neq -\mu_e$, so $\frac{\kappa}{2}(J-1)^2$ does not combine with the deviatoric term $\frac{\mu_e}{2}(\text{tr}(\mathbf{F}^T \mathbf{F}) - 2)$ to leave the reference configuration stress-free, and in practice it is used together with an isochoric (deviatoric/volumetric) split of $\mathbf{F}$. The point is that a standard-looking polynomial penalty already forfeits the benign tensile behaviour enjoyed by the logarithmic model (8) actually used in [8].

Remark 4 (Self-consistency of (NC)). Hypothesis (NC) is compatible with the singular kinematics (14): an angular profile Φ corresponding to opening/shear produces $\nabla \mathbf{u}$ whose dominant eigenvalue is positive (radial extension), so $\lambda_{\max} \rightarrow \infty$ while $\lambda_{\min} = \det \mathbf{F} / \lambda_{\max}$ need not approach zero. The complementary profile, in which the dominant eigenvalue of $\nabla \mathbf{u}$ is negative, drives $\lambda_{\min} \rightarrow 0$ and violates (NC); that is exactly Scenario B below.

3.6 Scenario B: the compressive singularity (the pathological case)

Now assume the far-field loading pushes the material *into* the corner ($K < 0$, radial compression).

Kinematic consequence. The naive linear prediction gives $\nabla \mathbf{u}$ a negative dominant eigenvalue scaling as $-|K|r^{\alpha-1}$. Because $\mathbf{F} = \mathbf{I} + \nabla \mathbf{u}$, as $r \rightarrow 0$ (or as the pseudo-time load factor t increases) there is a critical threshold where the local contraction annihilates the material volume:

$$\det \mathbf{F} \rightarrow 0 \implies J \rightarrow 0, \quad \lambda_{\min} \rightarrow 0. \quad (34)$$

Operator blow-up (and its model dependence). Here it is the *inverse channel* of (20) that fires: $\|\mathbf{F}^{-1}\| = \lambda_{\min}^{-1} \rightarrow \infty$. The dilation channel $\Theta(J)$ is now probed at $J \rightarrow 0^+$ rather than $J \rightarrow \infty$, and—exactly as in tension, but with the roles reversed—its behaviour discriminates between the two penalties. Writing $J \sim \lambda_{\min}$ near the corner (one large stretch, cf. §3.4),

$$\Theta_{\log}(J) \sim 2\lambda_e |\log J| \rightarrow \infty, \quad \Theta_{\text{quad}}(J) \sim 2\kappa J \rightarrow 0 \quad (J \rightarrow 0^+), \quad (35)$$

so that the tangent norm $\|\mathbb{L}\| \leq \mu_e + \Theta(J)\|\mathbf{F}^{-1}\|^2$ behaves as

$$\|\mathbb{L}(\mathbf{x})\| \sim \begin{cases} \mathcal{O}(\lambda_{\min}^{-2} \log \frac{1}{\lambda_{\min}}), & f = f_{\log}, \\ \mathcal{O}(\lambda_{\min}^{-1}), & f = f_{\text{quad}}. \end{cases} \quad (36)$$

Thus the logarithmic penalty—benign under tension—produces here the *stronger* blow-up, $\lambda_{\min}^{-2}$ *amplified* by a logarithm, whereas the quadratic penalty gives the milder rate $\lambda_{\min}^{-1}$. This reversal is not a redemption of the quadratic model: $f_{\text{quad}}(J) \rightarrow \frac{\kappa}{2} < \infty$ as $J \rightarrow 0^+$, so it provides *no energy barrier against volume collapse* and violates Assumption 1; the milder tangent growth merely reflects this missing barrier (the material is free to interpenetrate at finite energy), which is a worse pathology, not a better one. The logarithmic penalty’s stronger blow-up is, conversely, the analytic price of a genuine barrier $f_{\log}(J) \rightarrow +\infty$ —and it is precisely that barrier which forces, and gives physical meaning to, the bifurcation rescue of Section 4.

Ellipticity is *not* lost — the instability is structural. It is tempting to attribute the breakdown to a loss of strong ellipticity (an indefinite acoustic tensor). For this neo-Hookean model that is *false*, and it is worth stating the fact precisely, as it shapes the rest of the analysis.

Lemma 5 (Compressive strong ellipticity of the neo-Hookean tangent). *For the energy (7), the rank-one form of the tangent is exactly (18), $L_{iJkL}m_iN_Jm_kN_L = \mu_e|\mathbf{m}|^2|\mathbf{N}|^2 + f''(J)J^2(\mathbf{m} \cdot \mathbf{F}^{-T}\mathbf{N})^2$. Wherever $f''(J) \geq 0$ — in particular throughout compression, $J \leq 1$, for both penalties (8) — the tangent is strongly elliptic with constant $\geq \mu_e > 0$, uniformly in $\mathbf{F}$, and the acoustic tensor $Q_{ik}(\mathbf{N}) = L_{iJkL}N_JN_L$ is positive definite and never degenerates under compression.*

Proof. The identity is (18) (Lemma 1). As $J \rightarrow 0^+$, $f''(J)J^2 \rightarrow +\infty$ for $f_{\log}$ (cf. (52)) and $f''(J)J^2 = \kappa J^2 \geq 0$ for f_{quad} ; in both cases the second term only *adds* to the constant μ_e . $\square$ $\square$

Thus the compressive corner is *not* a material instability: the volumetric stiffening that blows up $\|\mathbb{L}\|$ is exactly the term $f''(J)J^2$ in (18), and it is positive — it strengthens ellipticity while destroying conditioning. What fails is the *boundedness* of the operator, not its positivity.

Conclusion for Scenario B. The assumption that the deformation smoothly follows the fundamental path into the corner therefore produces, not a loss of ellipticity, but two distinct effects: (i) a blow-up of the tangent *norm*, $\|\mathbb{L}\| \rightarrow \infty$ (eq. (36)), which renders the Newton step catastrophically ill-conditioned even though it remains elliptic; and (ii) for a barrier penalty $f_{\log}$, an unphysical infinite accumulation of energy ($f(J) \rightarrow \infty$ as $J \rightarrow 0$), signalling that the smooth fundamental path is not the physical solution. The genuine mechanism that intervenes is a *structural* bifurcation of the compressed state — the corner analogue of *creasing*, an instability of a material that remains everywhere strongly elliptic (Lemma 5), driven by the geometry and boundary conditions rather than by a change of type of the PDE. To capture it we introduce in Section 4 an imperfect perturbation that regularises the bifurcation and bounds the singularity; we work from here on with the barrier model $f_{\log}$, for which collapse is genuinely resisted.

4 Imperfect bifurcation analysis and asymptotic bounds

The previous section showed that the perfect compressive path drives $\lambda_{\min} \rightarrow 0$ and blows up the tangent norm while the material remains strongly elliptic (Lemma 5), so the breakdown is a structural instability rather than a loss of type. The premise of this section is that this scenario is an idealisation: a real system always carries small disturbances—geometric imperfections, load eccentricities, material inhomogeneity—and these select a nearby *regular* solution branch on which the collapse never quite happens. We model the disturbance by a single scalar amplitude ϵ , carry out an imperfect-bifurcation analysis, and quantify how the tangent blows up as $\epsilon \rightarrow 0$. The punchline is that the blow-up is confined to a vanishingly small neighbourhood of the corner and proceeds at a controlled algebraic rate, so that for every $\epsilon > 0$ the problem is regular.

4.1 The regularised problem and the fundamental path

We localise to the corner neighbourhood B_ρ of §3, regard the loading as transmitted through data on the outer arc $\partial B_\rho \setminus (\Gamma_D \cup \Gamma_N)$, and introduce a pseudo-time load parameter $t \in [0, 1]$ that monotonically scales the compressive part of that data. Following the answer to the question posed in Scenario B, we add a small, fixed volumetric body force $\epsilon \mathbf{p}(\mathbf{x})$ with $\|\mathbf{p}\|_{L^2(B_\rho)} = 1$ and $\epsilon \ll 1$ as the imperfection (we write the field as $\mathbf{p}$ to avoid collision with the penalty $f(J)$). The regularised equilibrium is $G(\mathbf{u}, t, \epsilon) = 0$,

$$\langle G(\mathbf{u}, t, \epsilon), \mathbf{v} \rangle = \int_{B_\rho} \mathbf{P}(\mathbf{F}) : \nabla \mathbf{v} \, dV - \epsilon \int_{B_\rho} \mathbf{p} \cdot \mathbf{v} \, dV - \ell_t(\mathbf{v}) = 0 \quad \forall \mathbf{v} \in V, \quad (37)$$

with $\mathbf{P} = \partial \Psi / \partial \mathbf{F}$, V the admissible test space, and ℓ_t the (linear) load functional at pseudo-time t . Let $\mathbf{u}_0(t)$ denote the *fundamental path*, i.e. the branch of (37) with $\epsilon = 0$ that continues the low-load compressive solution. We use the barrier penalty $f_{\log}$ throughout, so the collapse $J \rightarrow 0$ is genuinely resisted (Assumption 1); along $\mathbf{u}_0(t)$ the corner stretch $\lambda_c(t) := \lambda_{\min}(\mathbf{F}_0(t))$ decreases as $t \uparrow 1$ but, on the bifurcating branch we construct, never reaches zero.

Remark 5 (The instability is structural, not material). By Lemma 5 the tangent stays strongly elliptic throughout compression, so the bifurcation here is a *structural* one (a buckling of the boundary-value problem), not a material change of type. We therefore work with the global form

on the bounded domain B_ρ : the operator $\mathcal{L}(t) = \partial_{\mathbf{u}} G(\mathbf{u}_0(t), t, 0)$ is uniformly elliptic with compact resolvent, hence has discrete spectrum, and we assume (Assumption 6) that a simple eigenvalue first crosses zero at t_c . What makes the problem delicate is not a loss of ellipticity but the *norm* blow-up $\|\mathbb{L}\| \rightarrow \infty$ at the corner (eq. (36)): the coercivity constant stays positive while the boundedness constant diverges, so the crossing eigenvalue $\mu_1(t)$ and the conditioning are controlled by different mechanisms.

4.2 What is provable, and what must be assumed

It is worth separating the genuinely provable facts about the critical point from the irreducible hypotheses. Let $Q_t(\mathbf{v}) := \int_{B_\rho} \nabla \mathbf{v} : \mathbb{L}(\mathbf{F}_0(t)) : \nabla \mathbf{v} dV$ be the second-variation form on the fundamental path and $\mu_1(t) := \inf_{\|\mathbf{v}\|_V=1} Q_t(\mathbf{v})$.

The critical load exists (conditionally provable). A clarification is essential here. By Lemma 5 the neo-Hookean material remains strongly elliptic throughout compression, so $\mu_1(t)$ *cannot* be driven negative by a rank-one (material) instability: any crossing must be a *structural* loss of positivity of the global form Q_t , i.e. a buckling of the boundary-value problem in which the destabilising test field is not rank-one. This is the creasing mechanism.

Lemma 6 (Existence of the critical load). *Assume (i) coercivity at zero load, $\mu_1(0) > 0$, and (ii) a structural instability of the compressed state: there exist $t_* \in (0, 1)$ and $\mathbf{v}_* \in V$ with $Q_{t_*}(\mathbf{v}_*) < 0$. Then μ_1 is continuous on $[0, t_*]$ and there is a first $t_c \in (0, t_*]$ with $\mu_1(t_c) = 0$ and $\mu_1(t) > 0$ for $t < t_c$.*

Proof. At $t = 0$, $\mathbf{F}_0 = \mathbf{I}$ and $\mathbb{L} = \mu_e \mathbb{I} + (\text{volumetric at } J=1)$ is positive definite, so with Korn's inequality $\mu_1(0) > 0$. Hypothesis (ii) gives $\mu_1(t_*) \leq Q_{t_*}(\mathbf{v}_*) / \|\mathbf{v}_*\|_V^2 < 0$. As $t \mapsto \mathbb{L}(\mathbf{F}_0(t))$ is continuous, so is μ_1 , and the intermediate value theorem gives the first zero t_c . $\square$ $\square$

Hypothesis (i) is provable for neo-Hookean (positive-definiteness at $\mathbf{F} = \mathbf{I}$ plus Korn). Hypothesis (ii) — the existence of a structurally destabilising field — is the genuine content and is *not* elementary: it is the corner counterpart of the Biot/creasing instability [1, 3] and we do not establish it here.

The critical mode (assumed).

Assumption 6 (Critical mode). At t_c the infimum $\mu_1(t_c) = 0$ is attained at a mode $\phi \in V$, $\|\phi\|_V = 1$, spanning a one-dimensional, isolated kernel of $\mathcal{L}_c$ (the rest of the spectrum bounded away from 0), with ϕ localised near the corner.

This is the irreducible hypothesis: *attainment* (the minimising sequence does not escape into the corner as essential spectrum) and *simplicity*. Simplicity is generic and, for a symmetric corner, can be argued from a reflection symmetry about the bisector; discreteness is the borderline-elliptic difficulty already flagged in Remark 5. We do not prove it.

The crossing is transversal (provable under monotone loading).

Lemma 7 (Transversality). *Suppose the loading is monotonically destabilising at t_c ,*

$$\langle \phi, \dot{\mathbb{L}}(t_c) \phi \rangle := \int_{B_\rho} \nabla \phi : \partial_t \mathbb{L}(\mathbf{F}_0(t_c)) : \nabla \phi dV < 0. \quad (\text{M})$$

Then, under Assumption 6, μ_1 is differentiable at t_c with $a := -\mu_1'(t_c) = -\langle \phi, \dot{\mathbb{L}}(t_c) \phi \rangle > 0$.

Proof. For a simple eigenvalue, first-order perturbation theory gives $\mu'_1(t_c) = \langle \phi, \dot{\mathbb{L}}(t_c)\phi \rangle$; hypothesis (M) fixes the sign. $\square$ $\square$

Condition (M) is the precise meaning of “more compression $\Rightarrow$ less stiffness in the buckling direction” and holds for any monotone compressive load path; it is an input on the *loading*, not on the solution.

The imperfection is active (provable, generic).

Lemma 8 (Generic activation). *For every $\mathbf{p}$ outside the closed codimension-one subspace $\phi^\perp$, the projection $\eta := \int_{B_\rho} \mathbf{p} \cdot \phi \, dV \neq 0$. Hence $\eta \neq 0$ for generic imperfection.*

The bifurcation is subcritical (assumed). The one ingredient genuinely beyond reach of soft arguments is the *sign* of the nonlinearity. Guided by the creasing analogy [1, 3]—surface instabilities of compressed elastomers are well known to be subcritical—we assume it here rather than pretend to derive it.

Assumption 7 (Subcritical, non-degenerate reduced potential). The reduced bifurcation equation is subcritical with quintic saturation,

$$b := \frac{1}{6} \langle \phi, \partial_{\mathbf{u}}^3 G[\phi, \phi, \phi] \rangle < 0, \quad d := \frac{1}{120} \langle \phi, \partial_{\mathbf{u}}^5 G[\phi, \dots, \phi] \rangle > 0, \quad (38)$$

and the quadratic coefficient vanishes by the reflection symmetry $w \mapsto -w$ of the perfect compressive state about the corner bisector.

4.3 Lyapunov–Schmidt reduction

Proposition 1 (Normal form). *Under Assumption 6, Lemma 7 and Assumption 7, there is a neighbourhood of $(\mathbf{u}_0(t_c), t_c, 0)$ in which the solutions of (37) are exactly*

$$\mathbf{u}(t, \epsilon) = \mathbf{u}_0(t) + w \phi + \psi(w, t, \epsilon), \quad \psi \in V^\perp := \{\phi\}^\perp, \quad (39)$$

where ψ is smooth with $\psi = \mathcal{O}(w^2 + \epsilon)$, and the amplitude w solves the scalar imperfect bifurcation equation, retained to quintic order because the cubic is destabilising ($b < 0$),

$$a(t_c - t)w + b w^3 + d w^5 = \epsilon \eta + \mathcal{O}(\epsilon^2 + \epsilon w + |t_c - t| w^2 + w^6), \quad (40)$$

with $a > 0$ (Lemma 7), $b < 0 < d$ (Assumption 7) and η as in Lemma 8.

Proof. Let P be the V -orthogonal projection onto $\text{span}\{\phi\}$ and $Q = I - P$. Write $\mathbf{u} = \mathbf{u}_0(t) + w \phi + \psi$ with $\psi \in V^\perp$. The range equation $Q G = 0$ has invertible linearisation $Q \mathcal{L}_c Q$ on $V^\perp$ by Assumption 6, so the implicit function theorem yields a unique smooth $\psi = \psi(w, t, \epsilon)$ with $\psi(0, t_c, 0) = 0$ and $\psi = \mathcal{O}(w^2 + \epsilon + |t_c - t|)$. Substituting into the kernel equation $\langle G, \phi \rangle = 0$ and Taylor-expanding G about $(\mathbf{u}_0(t_c), t_c, 0)$ gives (40): the linear term $a(t_c - t)w$ from Lemma 7, the cubic $b w^3$ and quintic $d w^5$ from Assumption 7 (the quadratic vanishing by symmetry), and the inhomogeneity $\epsilon \eta$ from $-\epsilon \langle \mathbf{p}, \phi \rangle$. Because $b < 0$, the cubic alone does not saturate the amplitude, which is why the quintic $d w^5$ ($d > 0$) is retained. $\square$ $\square$

4.4 Subcritical branch structure and imperfection sensitivity

Perfect system. Set $\epsilon = 0$ and $t < t_c$. Dividing (40) by w gives $a(t_c - t) + bw^2 + dw^4 = 0$, i.e.

$$w^2 = \frac{-b \pm \sqrt{b^2 - 4ad(t_c - t)}}{2d}, \quad (41)$$

which (with $a > 0$, $b < 0$, $d > 0$) admits real nontrivial branches only for $t \geq t_L$, where

$$t_L := t_c - \frac{b^2}{4ad} < t_c, \quad w_L^2 = \frac{-b}{2d} = \mathcal{O}(1). \quad (42)$$

At t_L the two branches meet in a limit point (fold) at the *finite* amplitude w_L . The fundamental branch $w = 0$ is stable for $t < t_c$ and unstable for $t > t_c$; the upper nontrivial branch is stable. Hence on $[t_L, t_c)$ the compressed state and a finite-amplitude creased state coexist: the bifurcation is subcritical, with hysteresis and a *snap* at t_c .

Imperfect system.

Proposition 2 (Imperfection sensitivity, Koiter $\frac{2}{3}$ law). *For $\epsilon > 0$ the compressed branch terminates at a limit point that issues from the bifurcation point $(0, t_c)$ and is depressed below t_c according to*

$$t_s(\epsilon) = t_c - C |\eta \epsilon|^{2/3} + o(\epsilon^{2/3}), \quad C = \frac{3|b|^{1/3}}{2^{2/3}a} > 0, \quad (43)$$

so the load at which the compressed branch ceases to exist is reduced by $\mathcal{O}(\epsilon^{2/3})$. The turning amplitude there is itself small, $w_ = \mathcal{O}(\epsilon^{1/3})$; the $\mathcal{O}(1)$ amplitude is that of the creased branch onto which the structure then snaps.*

Proof. Write (40) as $g(w, t) = 0$ with

$$g(w, t) := a(t_c - t)w + bw^3 + dw^5 - \epsilon\eta,$$

the analytic remainder of (40) being subdominant at the scaling found below (we verify this at the end). A limit point in the load is a point of the solution set at which the branch turns, i.e. at which t is stationary along the curve; differentiating $g = 0$ implicitly, this is the pair

$$g(w, t) = 0, \quad g_w(w, t) = a(t_c - t) + 3bw^2 + 5dw^4 = 0. \quad (44)$$

We follow the limit point that issues from the bifurcation point $(0, t_c)$ as $\epsilon \rightarrow 0$; this is the one that caps the slightly deflected compressed branch and hence sets the snap load.

Turning amplitude. Using the second equation of (44) to eliminate the load, $a(t_c - t) = -(3bw^2 + 5dw^4)$, and substituting into the first gives a single equation for the turning amplitude,

$$-2bw^3 - 4dw^5 = \epsilon\eta. \quad (45)$$

Because $b < 0$, the small root of (45) as $\epsilon \rightarrow 0$ obeys $-2bw^3 = \epsilon\eta(1 + o(1))$, the quintic entering only at relative order w^2 ; hence

$$w_*(\epsilon) = \left(\frac{|\eta|\epsilon}{2|b|}\right)^{1/3} (1 + o(1)) = \mathcal{O}(\epsilon^{1/3}), \quad (46)$$

in agreement with the unstable-branch scaling (47) evaluated at the turn. In particular $w_* \rightarrow 0$, so at this limit point the cubic dominates the quintic, $dw_*^4/(bw_*^2) = \mathcal{O}(\epsilon^{2/3})$.

Snap load. Inserting (46) into $g_w = 0$ and discarding the quintic, a relative $\mathcal{O}(\epsilon^{2/3})$ correction,

$$a(t_c - t_s(\epsilon)) = -3b w_*^2 (1 + o(1)) = 3|b| \left(\frac{|\eta| \epsilon}{2|b|} \right)^{2/3} (1 + o(1)),$$

which is (43) with $C = 3|b|^{1/3}/(2^{2/3}a) > 0$, the classical Koiter $\frac{2}{3}$ power law [5, 2].

Remainder. At the limit point $w = \mathcal{O}(\epsilon^{1/3})$ and $t_c - t = \mathcal{O}(\epsilon^{2/3})$, so the omitted terms $\mathcal{O}(\epsilon^2 + \epsilon w + |t_c - t| w^2 + w^6)$ of (40) are $\mathcal{O}(\epsilon^{4/3})$, a relative $\mathcal{O}(\epsilon^{1/3})$ against the dominant balance $a(t_c - t)w \sim b w^3 \sim \epsilon \eta = \mathcal{O}(\epsilon)$. They perturb w_* and $t_s(\epsilon)$ only within the stated $o(\epsilon^{2/3})$, leaving the exponent and C unchanged. $\square$ $\square$

The unstable continuation branch. Separately, near t_c the small-amplitude (backward-bending, unstable) branch of (40) satisfies, at $t = t_c$, $b w^3 = \epsilon \eta (1 + o(1))$ with the quintic subdominant, so

$$|w(\epsilon)| = \left(\frac{|\eta|}{|b|} \right)^{1/3} \epsilon^{1/3} (1 + o(1)), \quad \epsilon \rightarrow 0. \quad (47)$$

This is the branch a displacement- or arc-length-controlled solver tracks through the instability, and it is the one that feeds the tangent estimate below; it is *not* the branch a load-controlled physical system follows.

4.5 Kinematic relief: lower bound on J

The heart of the regularisation is geometric: the buckling mode trades pure compression for rotation and shear, which lifts the determinant off zero. The reason the relief is second order in w —and hence why the scaling exponents below are what they are—is that the mode is, to leading order, a *rotation*, and rotations do not change volume at first order.

Lemma 9 (Determinant relief). *Let $\mathbf{F} = \mathbf{F}_0 + w \nabla \phi + \mathcal{O}(w^2)$ at the corner, with $\mathbf{F}_0 = \mathbf{F}_0(t_c)$, and let the rotation condition*

$$\text{cof } \mathbf{F}_0 : \nabla \phi = J_0 \mathbf{F}_0^{-T} : \nabla \phi = 0 \quad (48)$$

hold (the critical mode is volume-preserving at first order; this is established from the energy barrier in Lemma 10 below, exactly rather than assumed). With $d_\phi := \det(\nabla \phi) > 0$, in 2D,

$$J = \det \mathbf{F} = J_0 + w^2 d_\phi + \mathcal{O}(w^3), \quad (49)$$

so that, with $\lambda_c = \lambda_{\min}(\mathbf{F}_0)$ and $\lambda_{\max}(\mathbf{F}) = \mathcal{O}(1)$,

$$\lambda_{\min}(\mathbf{F}) = \frac{J}{\lambda_{\max}} \gtrsim C_1 \lambda_c + C_2 w^2, \quad C_1, C_2 > 0. \quad (50)$$

Proof. In 2D the determinant is quadratic: for any $\mathbf{A}, \mathbf{B}$, $\det(\mathbf{A} + w\mathbf{B}) = \det \mathbf{A} + w(\text{cof } \mathbf{A} : \mathbf{B}) + w^2 \det \mathbf{B}$. With $\mathbf{A} = \mathbf{F}_0$, $\mathbf{B} = \nabla \phi$, the first-order term is $\text{cof } \mathbf{F}_0 : \nabla \phi$, which vanishes by (48), leaving (49). Since $J = \lambda_{\min} \lambda_{\max}$ in 2D and $\lambda_{\max} = \mathcal{O}(1)$ on the relieved branch (the buckling redirects, but does not unboundedly stretch, the material near the corner), (49) gives (50). $\square$ $\square$

As the load approaches the instability the perfect corner stretch $\lambda_c \rightarrow 0$, so the w^2 term dominates the lower bound. On the *unstable continuation branch* (47) (the one a solver tracks through the instability), combining (50) with $w \sim \epsilon^{1/3}$,

$$\lambda_{\min}(\epsilon) \gtrsim C_2 w(\epsilon)^2 \sim \mathcal{O}(\epsilon^{2/3}), \quad \|\mathbf{F}^{-1}\| = \lambda_{\min}^{-1} \sim \mathcal{O}(\epsilon^{-2/3}). \quad (51)$$

On the stable physical branch the picture is different and milder; we contrast the two after Theorem 2.

The rotation condition (48) is not an extra assumption: it is *forced* by the energy barrier, as we now show. This is the precise content of the informal statement “buckling relieves compression”—the barrier makes the volume-preserving mode the only one of finite energy.

Lemma 10 (The barrier forces the rotation condition). *Let $f = f_{\log}$. Then the bulk modulus diverges at collapse,*

$$f''(J) = \frac{\mu_e + 2\lambda_e - 2\lambda_e \log J}{J^2} \sim \frac{2\lambda_e |\log J|}{J^2} \rightarrow +\infty \quad (J \rightarrow 0^+), \quad (52)$$

and consequently any mode ϕ of bounded second-variation energy $\langle \phi, \mathcal{L}\phi \rangle \leq C$ on the near-collapse fundamental state $\mathbf{F}_0$ satisfies

$$\text{cof } \mathbf{F}_0 : \nabla \phi = \mathcal{O}\left(\frac{J_0}{\sqrt{|\log J_0|}}\right) \rightarrow 0 \quad (J_0 \rightarrow 0^+). \quad (53)$$

In particular the critical buckling mode is first-order volume-preserving in the collapse limit, so (48) holds asymptotically along the loaded path, and the residual first-order term in (49) is dominated by the relief term $w^2 d_\phi$.

Proof. Differentiating $f'_{\log}(J) = -\mu_e/J + 2\lambda_e \log J/J$ gives (52). The volumetric part of the second variation of Ψ in the direction ϕ contains the term $f''(J_0)(\delta J)^2$, where $\delta J = \text{cof } \mathbf{F}_0 : \nabla \phi = J_0 \mathbf{F}_0^{-T} : \nabla \phi$ is the first variation of the determinant. Since every term other than this one is bounded on the fundamental state, boundedness of $\langle \phi, \mathcal{L}\phi \rangle$ forces $f''(J_0)(\delta J)^2 \leq C'$, hence $(\delta J)^2 \leq C'/f''(J_0) \sim C' J_0^2/(2\lambda_e |\log J_0|)$. Taking square roots yields (53). $\square$ $\square$

4.6 Regularised tangent bound and spatial localisation

Proposition 3 (Stable physical branch: bounded tangent). *Under load control the structure follows the compressed fundamental branch up to the snap at $t_s(\epsilon) \leq t_c$ (Proposition 2) and then jumps to the creased branch of amplitude $w = \mathcal{O}(1)$. On the compressed branch $\lambda_{\min} \geq \lambda_c(t) \geq \lambda_c(t_c) > 0$; after the snap the $\mathcal{O}(1)$ amplitude relieves the compression to $\lambda_{\min} = \mathcal{O}(1)$ via Lemma 9. Hence on the physical branch*

$$\max_{\mathbf{x} \in B_\rho} \|\mathbb{L}(\mathbf{x}, \epsilon)\| = \mathcal{O}(1) \quad \text{uniformly in } \epsilon, \quad (54)$$

and the sole effect of the imperfection is the $\mathcal{O}(\epsilon^{2/3})$ reduction (43) of the snap load. There is no tangent blow-up on the physical branch.

Theorem 2 (Unstable continuation branch: regularised blow-up). *Under Assumptions 1, 6, 7 and the conclusions (51) of Lemma 9, with the barrier penalty $f_{\log}$, the tangent along the unstable continuation branch (47)—the branch tracked by a displacement- or arc-length-controlled solver—satisfies*

$$\max_{\mathbf{x} \in B_\rho} \|\mathbb{L}(\mathbf{x}, \epsilon)\| \leq C \Theta(J) \|\mathbf{F}^{-1}\|^2 \sim \mathcal{O}(\epsilon^{-4/3} \log \frac{1}{\epsilon}), \quad \epsilon \rightarrow 0. \quad (55)$$

Proof. By (11), $\|\mathbb{L}\| \leq \mu_e + \Theta(J) \|\mathbf{F}^{-1}\|^2$. For $f_{\log}$, $\Theta(J) = \mathcal{O}(\log \frac{1}{J})$ as $J \rightarrow 0$ by (35), and on the unstable branch $J \sim \lambda_{\min} \sim \epsilon^{2/3}$ by (51), so $\Theta(J) = \mathcal{O}(\log \frac{1}{\epsilon})$; multiplying by $\|\mathbf{F}^{-1}\|^2 \sim \epsilon^{-4/3}$ gives (55). $\square$ $\square$

The two results together are the precise sense in which “the singularity never occurs in reality”: a load-controlled physical system snaps and keeps a bounded tangent (Proposition 3), and the divergent $\epsilon^{-4/3}$ rate is felt only by an algorithm that suppresses the snap and tracks the unstable branch through the instability (Theorem 2). The spatial localisation below applies to that branch.

The maximum is, however, achieved only in a shrinking zone, and we can now identify the localisation exponent exactly. The relevant geometry is the *collapse locus* of the perfect field.

Lemma 11 (Transversal collapse: $s = 1$). *On the perfect (unphysical) continuation of the fundamental path beyond t_c , let $\Sigma \subset B_\rho$ be the collapse locus on which the smallest stretch vanishes, $\lambda_{\min} = 0$ (equivalently $J = 0$). At any point of Σ where $\lambda_{\min}$ is a simple singular value of $\mathbf{F}$, $\lambda_{\min}$ is a smooth function of $\mathbf{x}$, and generically it vanishes transversally,*

$$\lambda_{\min}^{\text{field}}(\mathbf{x}) = \kappa_* \text{dist}(\mathbf{x}, \Sigma) + \mathcal{O}(\text{dist}^2), \quad \kappa_* > 0. \quad (56)$$

Hence the localisation exponent is $s = 1$. (A non-generic tangential contact, $\nabla \lambda_{\min} = 0$ on Σ , would give $s \geq 2$.)

Proof. Where the smallest singular value is simple it is an analytic function of the entries of $\mathbf{F}$, hence smooth in $\mathbf{x}$. A smooth scalar field vanishing on Σ has, by Taylor expansion, $\lambda_{\min} = \nabla \lambda_{\min} \cdot \mathbf{n} \text{dist}(\mathbf{x}, \Sigma) + \mathcal{O}(\text{dist}^2)$ with $\mathbf{n}$ the unit normal; transversality $\nabla \lambda_{\min} \neq 0$ (the generic case, by Sard’s theorem) gives (56) with $\kappa_* = |\nabla \lambda_{\min}| > 0$. $\square$ $\square$

On the imperfect branch the relief of (51) imposes the floor $\lambda_{\min}(\mathbf{x}, \epsilon) \gtrsim \max\{\kappa_* \text{dist}(\mathbf{x}, \Sigma), c \epsilon^{2/3}\}$, so the high-stiffness zone is the tube of half-width $\ell(\epsilon)$ around Σ in which the floor is active, with (using $s = 1$)

$$\ell(\epsilon) \sim \frac{c}{\kappa_*} \epsilon^{2/3} \xrightarrow{\epsilon \rightarrow 0} 0. \quad (57)$$

Proposition 4 (Narrowness of the singular zone). *With $\ell(\epsilon)$ as in (57) and $s = 1$ (Lemma 11), the set on which the tangent is near its maximum is a tube about Σ of area $\mathcal{O}(|\Sigma| \ell(\epsilon)) = \mathcal{O}(\epsilon^{2/3}) \rightarrow 0$, and the integrated stiffness obeys*

$$\int_{B_\rho} \|\mathbb{L}(\mathbf{x}, \epsilon)\|^q dV \leq C \left(1 + \epsilon^{-4q/3} \ell(\epsilon) \log^q \frac{1}{\epsilon}\right) = C \left(1 + \epsilon^{\frac{2}{3}(1-2q)} \log^q \frac{1}{\epsilon}\right), \quad (58)$$

which stays bounded as $\epsilon \rightarrow 0$ for every $q < \frac{1}{2}$. Thus, although the pointwise tangent diverges like $\epsilon^{-4/3}$, its L^q -mass over the corner remains finite for all $q < \frac{1}{2}$: the blow-up is genuinely localised to a tube of vanishing area, and for every $\epsilon > 0$ the regularised problem is well posed.

Proof. Split B_ρ into the tube $T_\epsilon = \{\text{dist}(\cdot, \Sigma) < \ell(\epsilon)\}$ and its complement. On T_ϵ , $\|\mathbb{L}\| \sim \epsilon^{-4/3} \log \frac{1}{\epsilon}$ (Theorem 2) and $|T_\epsilon| \sim |\Sigma| \ell(\epsilon) \sim \epsilon^{2/3}$, giving the displayed contribution. Off T_ϵ , using transversal coordinates $d = \text{dist}(\cdot, \Sigma)$ and $\lambda_{\min} \sim \kappa_* d$, one has $\|\mathbb{L}\| \sim d^{-2} \log \frac{1}{d}$, and $\int_{\ell(\epsilon)}^\rho d^{-2q} \log^q \frac{1}{d} dd$ converges as $\ell(\epsilon) \rightarrow 0$ precisely when $2q < 1$. Both regimes are controlled iff $q < \frac{1}{2}$. $\square$ $\square$

4.7 Discussion

The structure of the result is the one anticipated physically, and it splits cleanly along the stability of the branch followed. The idealised noiseless problem ($\epsilon = 0$) is singular—the tangent is unbounded and ellipticity is lost—but it is a measure-zero idealisation. Because the corner instability is subcritical (the creasing analogy, Assumption 7), a load-controlled physical system never traverses the singular continuation: it stays on the compressed branch and *snaps* to a finite-amplitude

crease at a load reduced by only $\mathcal{O}(\epsilon^{2/3})$ (Proposition 2), keeping a bounded tangent throughout (Proposition 3). The much-discussed divergence is therefore not seen by the physical system at all; it is the worst case for an algorithm—displacement or arc-length control, numerical continuation—that suppresses the snap and tracks the *unstable* branch through the instability. On that branch the determinant is still lifted off zero (Lemma 9, with the volume-preserving character of the mode *forced* by the energy barrier, Lemma 10), the tangent diverges only algebraically, $\epsilon^{-4/3}$ up to a logarithm (Theorem 2), and the singular character is confined to a tube of width $\ell(\epsilon) \sim \epsilon^{2/3} \rightarrow 0$ about the collapse locus, remaining L^q -integrable for all $q < \frac{1}{2}$ (Lemma 11, Proposition 4). That algebraic rate is the mirror image—through the inverse channel—of the benign tensile growth of §3.4; the same logarithm, traceable to the barrier $f_{\log}$, appears in both. Two practical consequences follow. First, the bifurcation is the corner-localised analogue of the classical *creasing* instability of compressed elastomers [1, 3], now seated at a geometric Dirichlet–Neumann junction rather than on a flat surface. Second, the dichotomy is a concrete warning for computation: a solver that continues through the instability will see the conditioning degrade as $\epsilon^{-4/3}$ and should either admit the snap or refine inside the tube (57); the load-control reduction (43) tells it when the snap is due. The reduction to scalar imperfect-bifurcation form follows the classical theory [2, 4, 5].

5 Conclusions

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